ANNAMALAI UNIVERSITY

FACULTY OF ENGINEERING AND TECHNOLOGY

ENGINEERING MATHEMATICS

B.E. - FOURTH SEMESTER

(Common to CS, IT and AI&ML)

II- Mid Term Test- May-2022

COURSE: **CSBS401/ITBS401/AIBS401- DISCRETE MATHEMATICS**

Time: 90 Minutes Max: 45 Marks

PART- A (6x 1 =6 Marks)

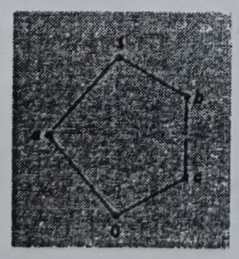
Answer ALL Questions

		Marks	CO	PO	BTL
1.	State the modular inequality of a Lattice	1	CO3	3,5	K2
2.	Define Boolean algebra.	1	CO3	1,2	K1
3.	Define a Monoid with an example	1	CO4	3,4	K1
4.	What is the identity element of the algebraic system $\{s,*\}$ Where S is the set of integers and * is defined by $a * b = a + b + 2$ for all $a, b \in S$.	1	CO4	1,2	K1
5.	Define regular graph with an example	1	CO5	2,3	K1
6	Define graph isomorphism	1	CO5	1,2	K2

PART- B (3x3=9 Marks)

Answer either (a) or (b) in each Question

7.	a)	Draw the Hasse diagram for the divisibility relation on the set D_{24}	3	co3	1,2	k2
		OR				
	b)	Verify whether the lattice given by the Hasse diagram is distributive	3	CO3	2,3	K2



8.	a)	show that in a group G, $(a * b)^{-1} = b^{-1} * a^{-1}$, for all $a, b \in G$	3	CO4	1,2	K3
		OR				

$\frac{32}{2318} + \frac{30}{2317} + \frac{28}{2316} + \frac{32}{2315}$ $\frac{36}{2313} + \frac{15}{2312}$

	b)	Show that $\{(1,2,3,4,5,6),\times_7\}$ is a cyclic group. .	3	CO4	3,4	K3
9.	a)	Explain Konigsberg bridge problem. Represent the problem by means of graph. Does the problem have a solution?	3	CO5	1,2	K3
	b)	Define Bipartite graph. Draw $K_{2,3}$ and $K_{3,3}$ graphs	3	CO5	2,3	K3

PART- C (10x3=30 Marks)

Answer either (a) or (b) in each Question

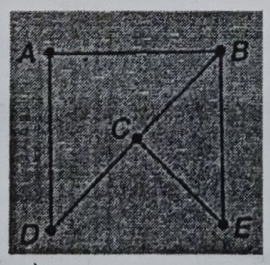
10	a)	Show that in a distributive lattice $\{L,\vee,\wedge\}$ if an element $a \in L$ has a complement then it is unique	10	CO4	2,3	K4

OR

	b)	Simplify the Boolean function $f(a,b,c,d) = \sum(0,1,2,3,4,6,7,8,9,11,15)$	10	CO4	2,3	K5
11	a)	show that $\{s_3,*\}$, where $s = \{1,2,3\}$ is a group under the operation of right composition of permutation	10	CO5	2,3	K4

OR

	b)	List the code words generated by the encoding function $e: B^3 \rightarrow B^6$ whose parity check matrix is $H = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	10	CO5	2,3	K5
12	a)	Summarize all the paths of length 4 between D and E for the following graph using adjacency matrix.	10	CO3	2,3	K5



OR

	b)	Find the shortest path from the vertex A to the vertex F	10	CO3	2,3	K5

